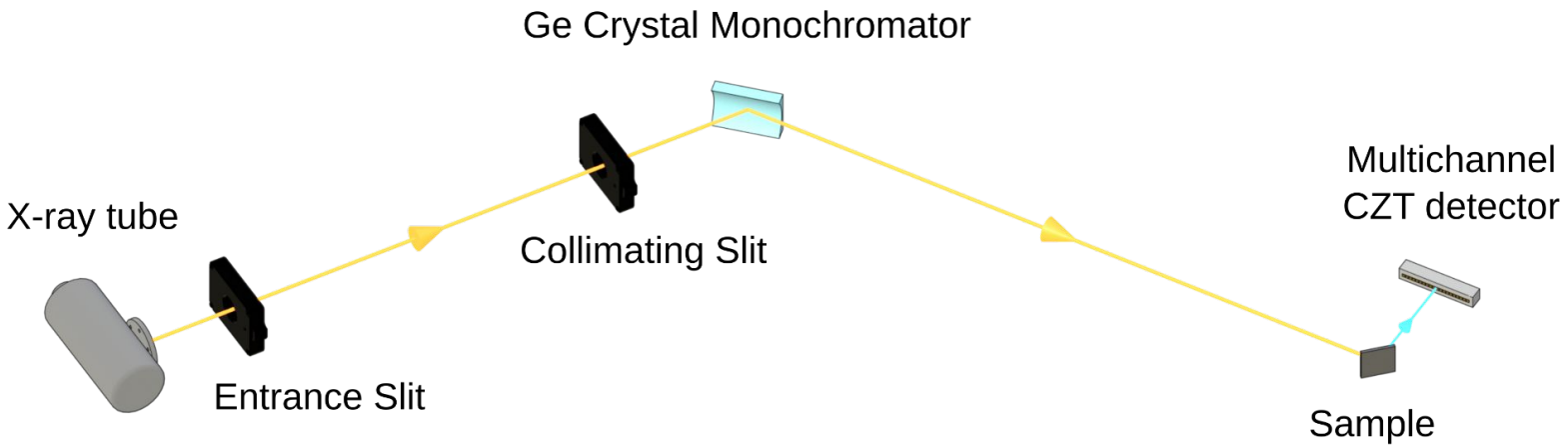
PRISM

Precision Room-temperature Instrumentation for high-energy X-ray absorption Spectroscopy Measurements

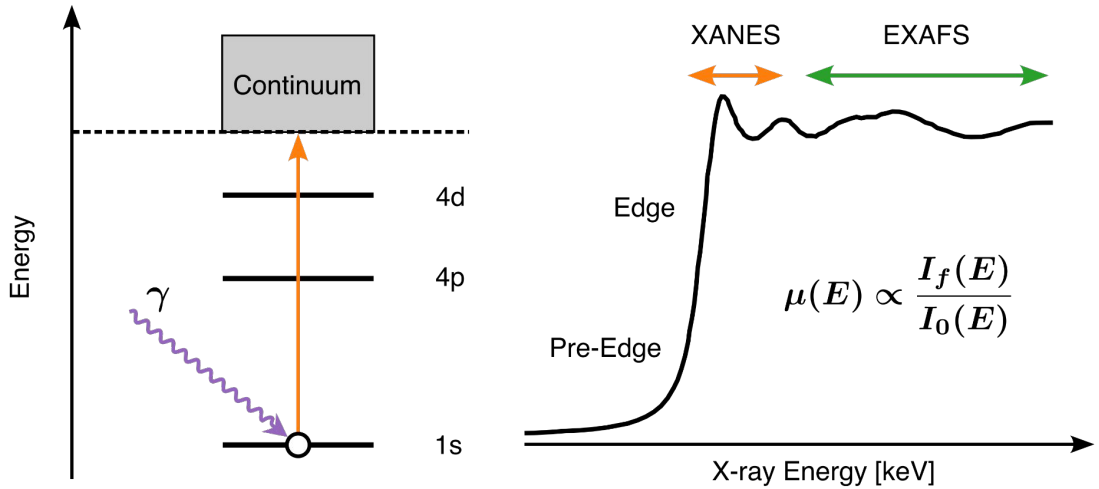
Responsabile di Progetto: Dr. Simone Manti, INFN - LNF
Referente di Progetto: Dr.ssa Catalina Curceanu, INFN - LNF

How can we perform laboratory XAS above 20 keV?



X-ray Absorption Spectroscopy (XAS): Chemical State and Local Structure

XAS probes element-specific **Absorption Edges** to reveal **Chemical State** and the **Local Atomic Environment**.



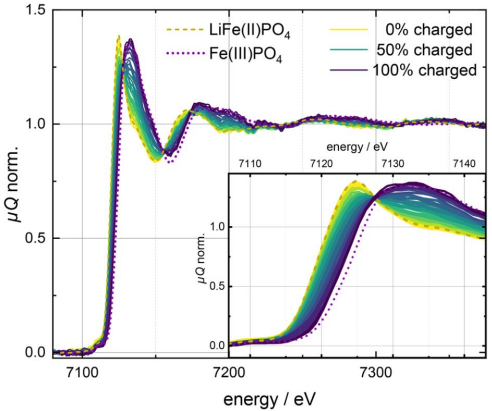
1. **Element selectivity**
   Core-level edges identify the absorbing element
2. **Pre-edge**
   Electronic states and local symmetry
3. **XANES**
   Oxidation state and coordination geometry
4. **EXAFS**
   Neighbor distances, coordination and disorder

State-of-the-Art XAS under Operating Conditions

Element-specific insight into **Electronic State** and **Local Atomic Structure**.

Laboratory Operando XANES

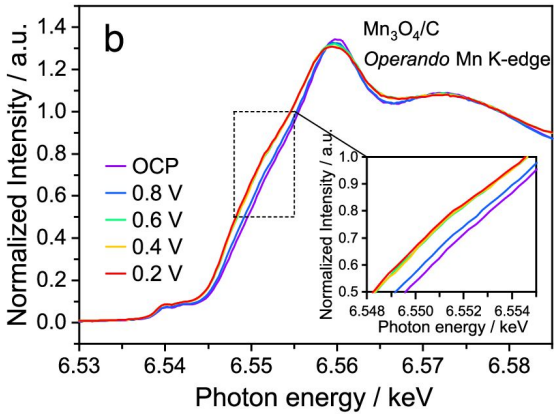
Lab XANES tracks $LiFePO_4$ conversion
Fe K-edge ~7.1 keV, 15 min per spectrum



Praetz et al., JAAS 40, 2447–2461 (2025), Fig. 2

Working Catalysts

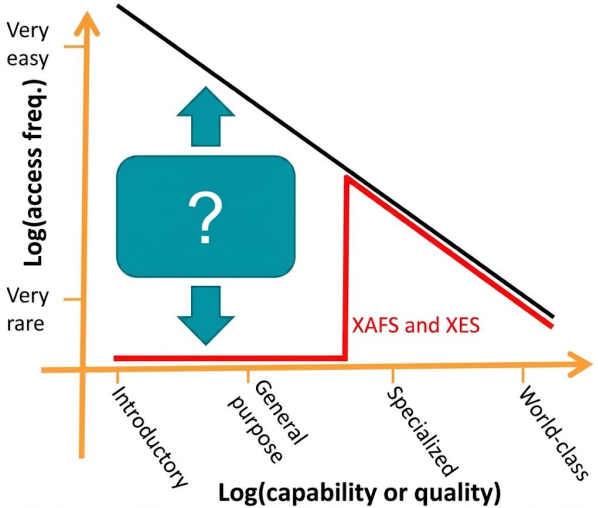
Operando XANES probes the **Mn K-edge ~6.5 keV** state under fuel-cell operating conditions



Li et al., Nature Communications (2025), Fig. 3b

Why bring XAS into the Laboratory?

- **Synchrotrons** and **FEL** remain the reference for XAS performance.
- **Laboratory instruments** with X-ray tubes offer routine access and flexible measurement times.



Jerry Seidler, et al. (2024)

- **Routine screening** and **repeated measurements**
- Long **In-Situ / Operando** campaigns
- Sample and protocol optimization for beamtime

Commercial Laboratory XAS already exists, but elevated cost and up to 20 keV!



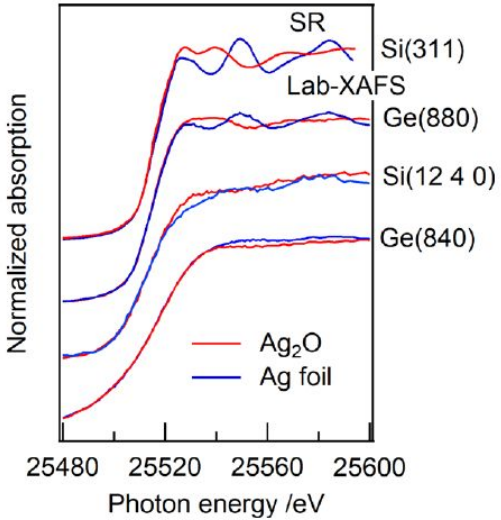
Sigray

≈ 1.5 M€



Laboratory XAS above 20 keV: Opportunities and Challenges

- K-edges of Mo, Pd, Ag and Sn are relevant to **Catalysis** and **Energy Materials**.
- **Closely spaced L-edges** can restrict the accessible EXAFS range.



Feasibility demonstrated in transmission

- Ge(880): ≈12 h for Ag K-edge XANES
- Ag K-edge EXAFS impractical at this count rate

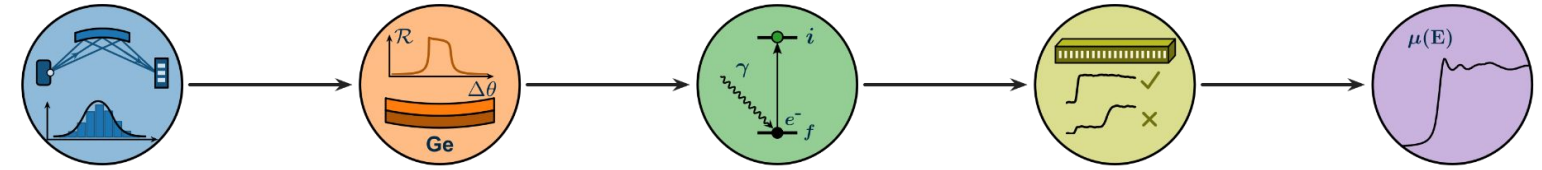
Spectral purity is also critical

- Al filters suppress lower-order reflections (440) and characteristic Ge K lines.

Challenge: quantitative fluorescence XAS with efficient detection and sufficient SNR in a single lab day!

Yamamoto. "Selection of Monochromator Crystal for Pb L- and Ag K-Edge XANES Using a Laboratory-Based Spectrometer." *X-Ray Spectrometry* (2026).

PRISM



SIMULATION
Ray tracing
Detector response

CRYSTAL
Bent Ge optics
Energy selection

XAS
X-ray absorption
spectroscopy

CZT
Multichannel detection
Pulse-shape analysis

XANES
Near-edge
structure

MAIN GOAL:

"To develop and validate a room-temperature laboratory platform for fluorescence-mode XAS above 20 keV, combining a high-resolution crystal monochromator with a custom multichannel CZT detector "

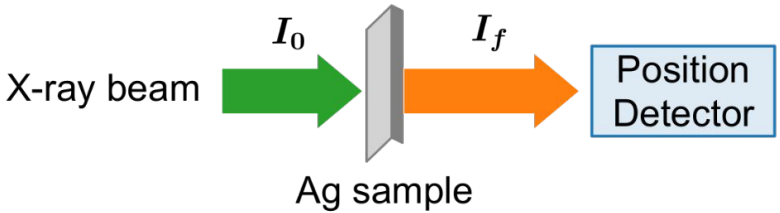
PRISM probes XAS K-edge with the characteristic Fluorescence line

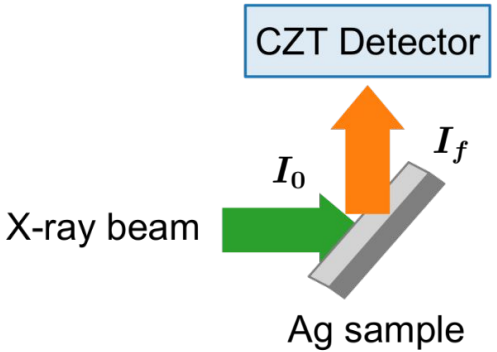
Transmission Mode

Requires sufficient transmission and a uniform sample thickness.

Fluorescence Mode

Enables measurements on dilute samples and samples unsuitable for transmission.

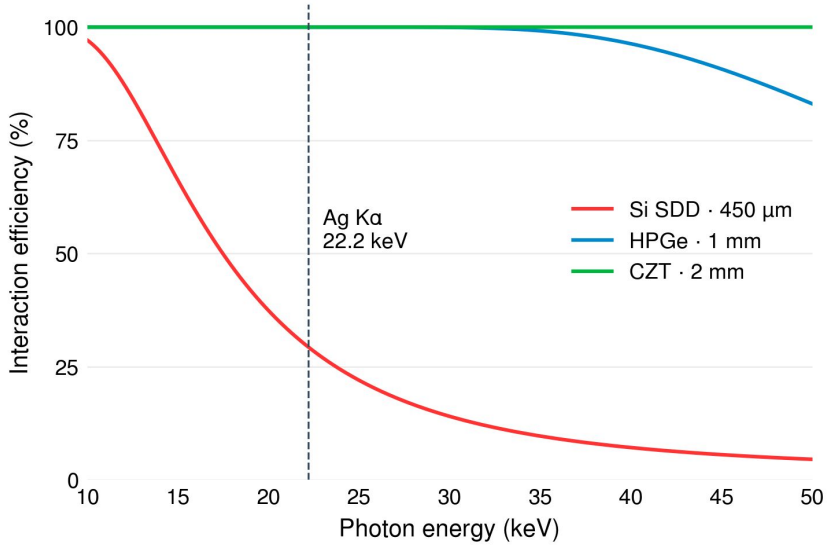




PRISM requires efficient, energy-resolving detection of weak fluorescence above 20 keV!

CZT have High Efficiency > 20 keV with Room-Temperature Operation

Fluorescence-Mode XAS above 20 keV requires high detection efficiency



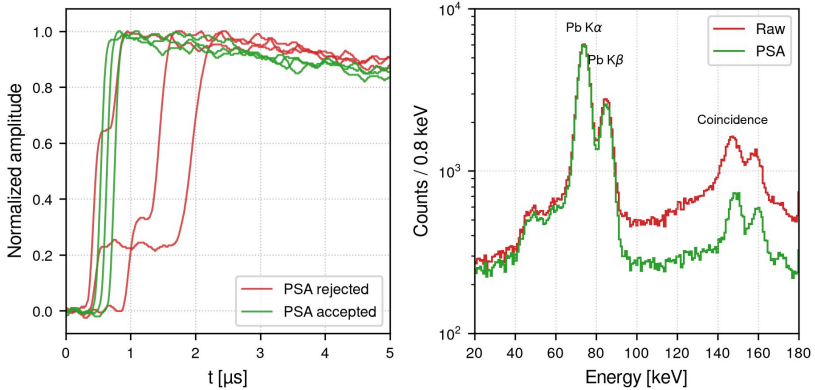
- **Efficient Fluorescence collection**
  Higher absorption at Ag Kα (22.2 keV) than **Silicon Drift Detectors (SDDs)**
- **Spectroscopy at Room-Temperature**
  No cryogenic cooling, unlike **High-Purity Germanium (HPGe)** detectors
- **My CZT experience at DAΦNE**
  High-flux Kaonic Atoms Hard-X-ray Spectroscopy with SIDDHARTA-2
  Artibani F., Manti S, et al., J. Phys. G (2026)

CZT Detector, Front-End Electronics and Waveform DAQ ~33% of PRISM's cost

A custom CZT detection chain for fluorescence XAS

- **Optimize fluorescence collection**
  A custom 20-pixel linear array with 500 μm pitch, tailored to the collection geometry.
- **Custom readout for varying count rates**
  Below/Above Edge, with /without (I_0) Target
- **Experience in Pulse Shape Analysis**
  - **High-Flux CZT@BTF:** Manti S., et al., Condensed Matter 11, 38 (2026).
  - **Low-Flux BEGe@LNGS:** Manti S., et al. JINST 21.01 (2026): P01036.

Building on prior CZT results at DAΦNE-BTF

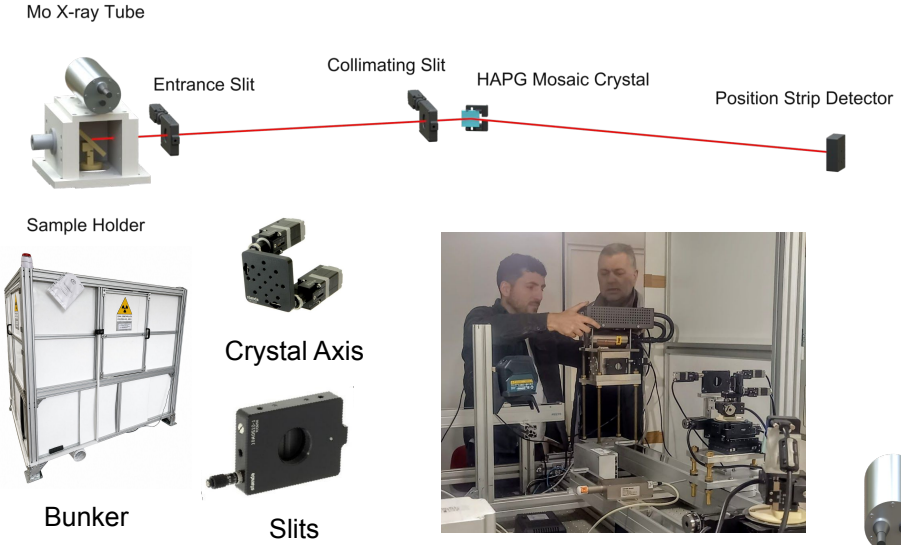


Pulse Shape Analysis suppresses **Coincidence Structures** and maintain **Pb Kα and Kβ lines**

M15 targets: FWHM/E ≤ 2% at Ag Kα, Dead Time < 10%, Pile-up contamination < 5%

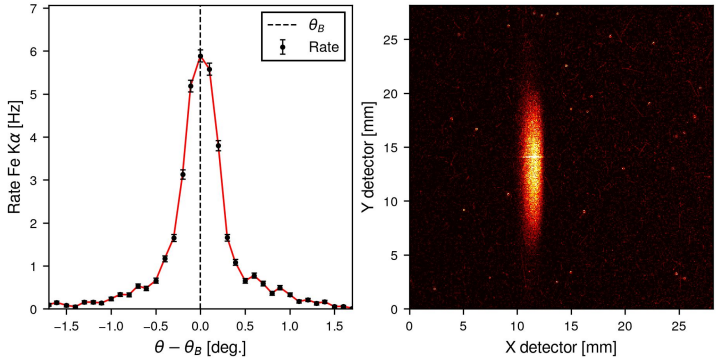
VOXES as the experimental basis for PRISM

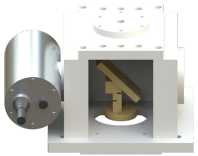
Existing VOXES platform (Scordo CSN5 2015)



[Manti et al., Condensed Matter 9.1 (2024): 19.]

Demonstrated **Angular Scan and Beam Focusing** (e.g. FeKα line)

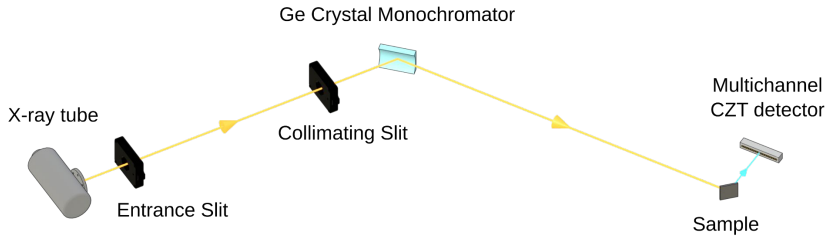




Recent upgrades enable **Absorption Measurements**

[Manti et al.. Engineering Innovations 19 (2026): 23-29.]

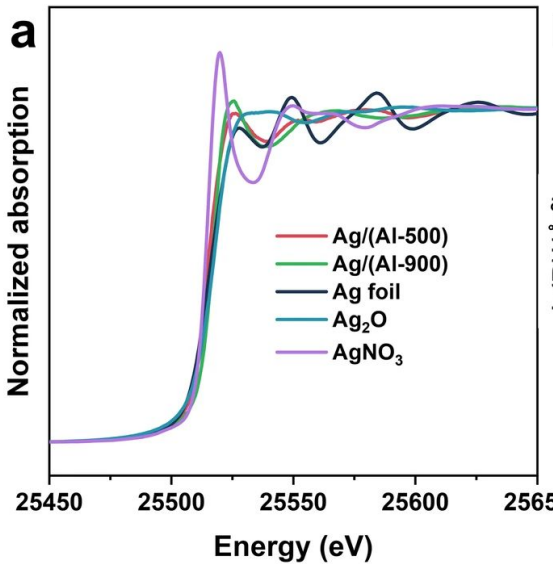
Scanning above 20 keV requires tailored Optics, Source and Motors



- **Ge Crystal · €20k**
  Narrow-band incident-energy selection for XANES: target ≤10 eV FWHM at the Ag K-edge. Ge(880) is the baseline candidate.
- **W-Anode X-ray tube · €17k**
  Higher continuum flux above 20 keV to provide sufficient photons after monochromatization.
- **Motorized Stages · €10k**
  Two rotations and one translation to switch reproducibly between fluorescence I_f and incident-flux I_0 measurements at each energy point.

RAY-TRACING + GEANT4 End-End simulations will define the final setup specifications

Ag K-edge XANES will be the decisive validation of PRISM



Li, Jiaxin, et al. *Comm.* 15.1 (2024): 3874.

- **REFERENCE SAMPLES**
  Ag foil + two Ag(I) compounds
  Different local coordination environments
- **PERFORMANCE TARGETS**
  ≤10 eV FWHM incident-beam bandwidth ≥20 edge-step SNR
- **VALIDATION**
  Energy calibration and reproducibility
  Comparison with reference XANES spectra

Work Plan and Milestones for PRISM

Work Packages (WP) for Simulation, CZT, Commissioning and Validation and **Milestones (MS)**

WORK PACKAGE (WP)	PLANNED ACTIVITY (A)	2027				2028			
		1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24
WP1 - Requirements, simulations and design freeze	A1.1 - Requirements definition								
	A1.2 - SHADOW4 ray-tracing optimization								
	A1.3 - GEANT4 fluorescence simulations								
	A1.4 - Final geometry selection			MS1					
WP2 - Room-temperature multichannel CZT spectroscopy for hard X-ray fluorescence XAS	A2.1 - CZT detector and front-end design								
	A2.2 - Multichannel DAQ and signal-processing development								
	A2.3 - CZT calibration and performance characterization								
	A2.4 - Detector validation and operating-point selection					MS2			
WP3 - Instrument integration and commissioning	A3.1 - Procurement of source, optics and motion stages								
	A3.2 - Mechanical design and spectrometer assembly								
	A3.3 - Hardware, software and control-system integration								
	A3.4 - Instrument commissioning and verification						MS3		
WP4 - Ag reference samples and realistic sample demonstration	A4.1 - Ag foil calibration and reproducibility tests								
	A4.2 - XANES validation on Ag reference compounds								
	A4.3 - Fluorescence XAS demonstration on a realistic sample								
	A4.4 - Performance characterization and operating limits								MS4

- **M6 Setup specifications frozen from Simulations**
- **M15 CZT spectroscopic performance demonstrated**
- **M18 Integrated instrument commissioned**
- **M24 Ag XANES demonstration Validated**

Organization and Synergies supporting PRISM

1.7 FTE at LNF, with specialist expertise for CZT, Crystal Optics and Simulations

Defined Responsibilities across WP1–WP4

Name	Unit	FTE	Responsibility/WP
Simone Manti	INFN-LNF	1.0	Project Leader; WP1–WP4
Alessandro Scordo	INFN-LNF	0.3	Crystal optics, ray tracing, integration; WP1/WP3
Massimiliano Bazzi	INFN-LNF	0.3	Electronics and DAQ; WP2
Catalina Curceanu	INFN-LNF	0.1	Scientific Supervisor and application validation; WP4
Total		**1.7**	-

- **VOXES:** equipment and test infrastructure.
- **Palermo/KAONNIS (CSN3):** CZT and calibration
- **VIP (CSN3):** pulse-shape analysis.
- **Further links:** SPHINX and EuAPS/AQUA

ENRICH COST Action

I am **co-leader of ENRICH Working Group 3** with M. Sánchez del Río, main developer of the Ray-Tracing software **SHADOW4**.



Mission at ESRF (Grenoble)

Letter of support for PRISM
Potential technical exchanges on CZT sensors and testing (WP2).

Budget for the integrated PRISM instrument

Item	Year	Cost (k€)
CZT detector and front-end	2027	30
Digitizer and DAQ	2027	23
Bent Ge crystal optics	2028	20
Motorized stages and controllers	2027	10
Cooled W-anode X-ray tube	2028	10
X-ray tube power supply	2028	7
Oral conference travel	2028	4
Ag reference targets	2027	2
Custom mechanical supports	2027	2
DataCloud CPU and storage	2027	2
Additional reference targets	2028	2
Total Requested	**2027–28**	**112**

112 k€

Requested budget
2027: 69 k€
2028: 43 k€

100 k€ in equipment
89% of the request

45 k€

Estimated in-kind value
VOXES tube and bunker
Motion and control systems

Expected Impact of the PRISM project

Fluorescence XAS for **Catalysis** and **Energy Materials** above 20 keV (Mo,Pd,Ag,Sn)

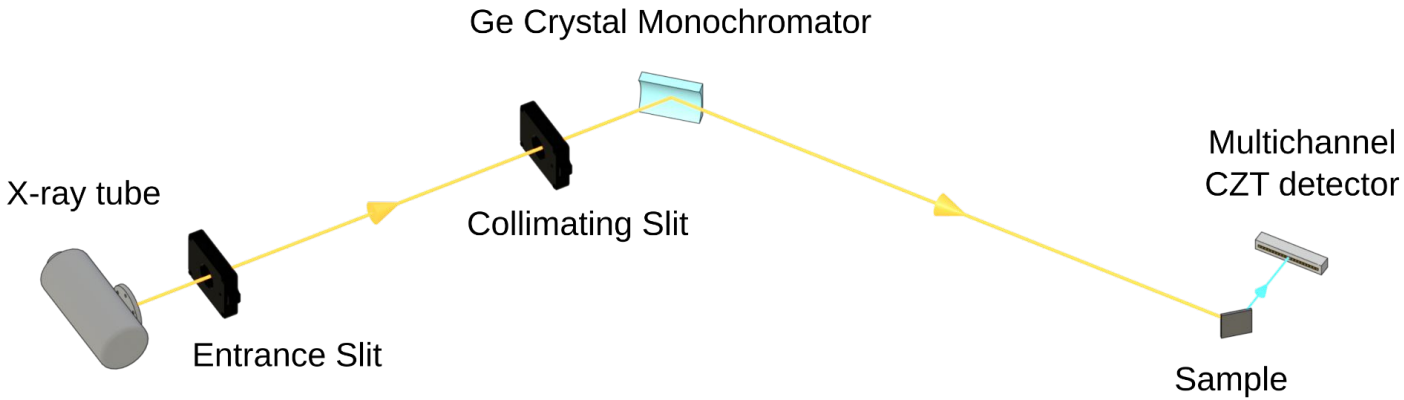
Scientific Access	Screen **Thick** or **Dilute Samples** and track slow changes. Optimize protocols before **Synchrotron Beamtime**.
Industrial Interest	**Rigaku** and **Quantum Design** support PRISM (letters) Quantum Design offers Sigray laboratory XAS systems.
INFN Capabilities	Reuse **CZT readout**, simulations and calibration. Future INFN capability (EuAPS,EuPRAXIA@AQUA,VIP, KAONNIS@JPARC)

Main Risks and practical Mitigations

Risk	Activities	Mitigation	PNR
R1. Limited accuracy of the end-to-end simulation	A1.2–A1.4	Benchmark against analytical calculations and VOXES data. Consult SHADOW4 developer M. Sánchez del Río through ENRICH WG3.	**Medium**
R2. Extended CZT optimization phase	A2.1–A2.4	Use the VOXES Si PIN for preliminary calibration at reduced efficiency while CZT optimization continues.	**Low**
R3. Delayed delivery of critical components	A3.1–A3.4	Start integration and alignment with compatible VOXES components, including non-optimized crystals.	**Medium**
R4. Lower than expected fluorescence count rate or SNR	A3.3–A4.4	Optimize scan steps, dwell time, acceptance and detector shaping. Preserve energy resolution and background rejection.	**High**

PRISM: fluorescence XAS above 20 keV at LNF

Goal: quantitative Ag K-edge XANES, building on VOXES



M24
Ag XANES validation

≤10 eV
Target beam FWHM

≥20
Target edge-step SNR

112 k€
Requested budget

Thank you for your Attention!

References

[1] Yamamoto. "Selection of Monochromator Crystal for Pb L- and Ag K-Edge XANES Using a Laboratory-Based Spectrometer." X-Ray Spectrometry (2026).

[2] Manti, Simone, and Alessandro Scordo. "Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies." Engineering Innovations 19 (2026): 23-29.

[3] Scordo, A., et al. "VOXES: a high precision X-ray spectrometer for diffused sources with HAPG crystals in the 2–20 keV range." Journal of Instrumentation 13.04 (2018): C04002-C04002.

[4] Manti, Simone, et al. "Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations." Condensed Matter 9.1 (2024): 19.

[5] Manti, Simone, et al. "Machine-learning-enhanced event selection for BEGe detectors in the VIP experiment." Journal of Instrumentation 21.01 (2026): P01036.

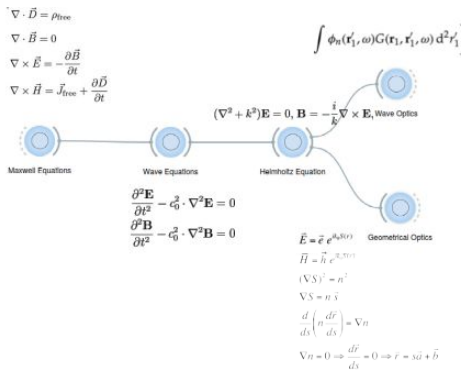
[6] Manti, Simone, et al. "Pulse-Shape Discrimination with Machine Learning for CZT Detectors at DAΦNE Beam Test Facility." Condensed Matter (2026).

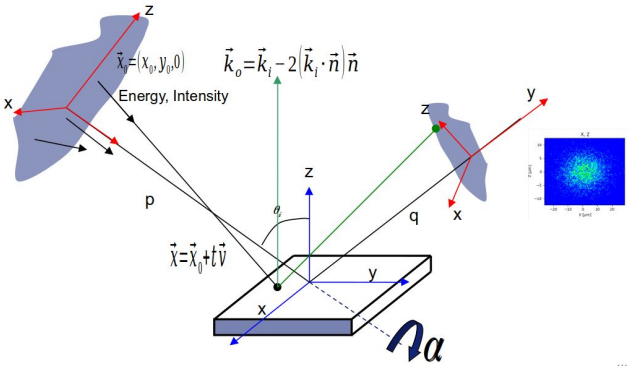
[7] Artibani, Francesco, et al. "Kaonic copper and fluorine absolute yields measurement with a CZT-based detection system at DAΦNE." Journal of Physics G: Nuclear and Particle Physics 53.9 (2026): 095102.

SPARE

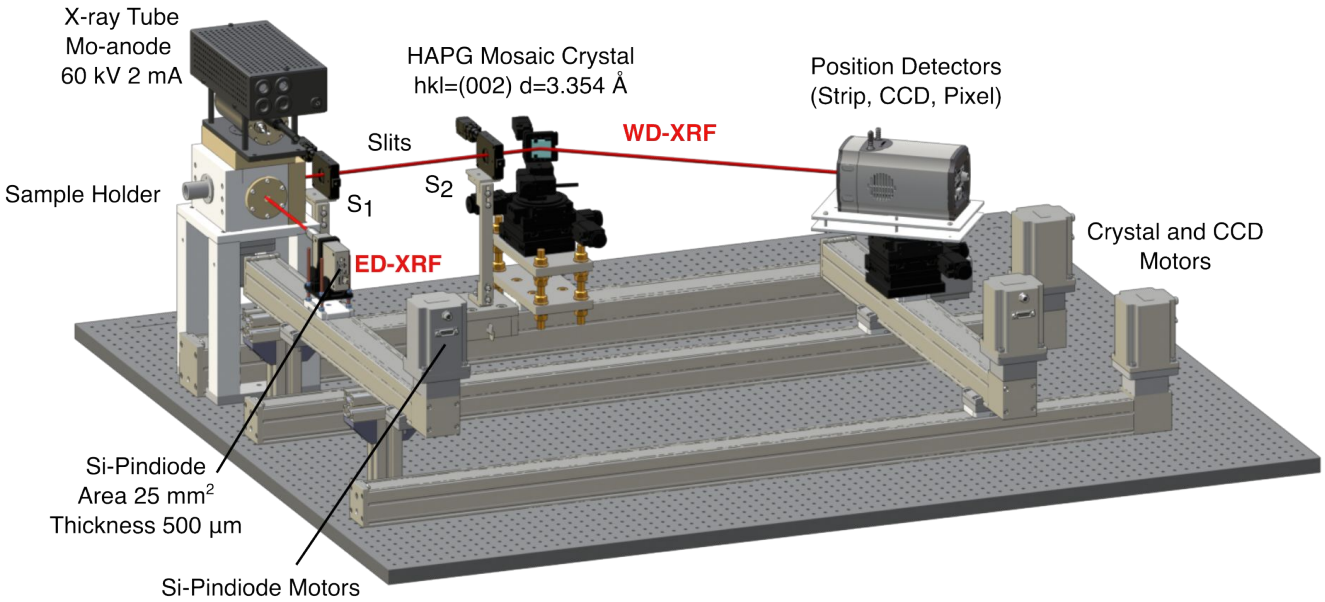
Ray-Tracing Simulations with SHADOW4

- Ray tracing in X-ray optics is a simulation method used to model how X-rays propagate and interact with optical components, like mirrors, lenses, and monochromators.
- It relies on approximate solutions to Maxwell's equations that are valid as long as the light waves propagate through and around objects whose dimensions are much greater than the light's wavelength.
- It's essential for designing and optimizing beamlines at synchrotrons as well as any X-ray optical system.





VOXES


X-ray Tube
Mo-anode
60 kV 2 mA
HAPG Mosaic Crystal
hkl=(002) d=3.354 Å
Position Detectors
(Strip, CCD, Pixel)
Slits
WD-XRF
Sample Holder
S1
S2
ED-XRF
Crystal and CCD
Motors
Si-Pindiode
Area 25 mm2
Thickness 500 µm
Si-Pindiode Motors